Kinematics

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Displacement, velocity, and acceleration**

A particle moves with displacement $s = t^3 - 6t^2 + 9t$ (in meters, t in seconds). Find expressions for its velocity v and acceleration a.

$v = \frac{ds}{dt} = 3t^2 - 12t + 9$. $a = \frac{dv}{dt} = 6t - 12$.

2 **[Foundational]** • **Target: Displacement, velocity, and acceleration**

A particle moves with displacement $s = 2t^3 - 9t^2 + 12t$ (in meters, t in seconds). Find its velocity at $t = 2$ seconds.

$v = \frac{ds}{dt} = 6t^2 - 18t + 12$. At $t = 2$: $v = 6(4) - 18(2) + 12 = 24 - 36 + 12 = 0$ m/s.

3 **[Foundational]** • **Target: The suvat equations for constant acceleration**

A car starts from rest and accelerates uniformly at 2.5 m/s^2 for 8 seconds. Find its final velocity and the distance traveled, using $v = u + at$ and $s = ut + \frac{1}{2}at^2$.

$v = u + at = 0 + 2.5(8) = 20$ m/s. $s = ut + \frac{1}{2}at^2 = 0 + \frac{1}{2}(2.5)(64) = 80$ m.

4 **[Foundational]** • **Target: The suvat equations for constant acceleration**

A ball is thrown upward with initial velocity $u = 15$ m/s. Using $g = 9.8$ m/s^2 (deceleration), find the maximum height reached, using $v^2 = u^2 + 2as$ with $v = 0$ at the top.

$0 = 15^2 + 2(-9.8)s$, so $s = \frac{225}{19.6} \approx 11.48$ m.

5 **[Foundational]** • **Target: The suvat equations for constant acceleration**

A cyclist decelerates uniformly from 12 m/s to rest over a distance of 36 meters. Find the deceleration, using $v^2 = u^2 + 2as$.

$v^2 = u^2 + 2as$: $0 = 12^2 + 2a(36) = 144 + 72a$, so $a = -2$ m/s^2 (a deceleration of 2 m/s^2).

6 **[Foundational]** • **Target: Velocity-time graphs**

A particle accelerates uniformly from $u = 2$ m/s to $v = 14$ m/s over 6 seconds, then travels at constant velocity for a further 4 seconds. The velocity-time graph is shown. Find the total distance traveled, using the area under the graph.

Distance in first phase (trapezoid): $\frac{1}{2}(2 + 14)(6) = 48$ m. Distance in second phase (rectangle): $14 \times 4 = 56$ m. Total: $48 + 56 = 104$ m.

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7 [Foundational] • Target: Velocity-time graphs

A particle decelerates uniformly from $v = 20$ m/s to rest over 5 seconds, giving a velocity-time graph that is a straight line from $(0, 20)$ to $(5, 0)$. Find the distance traveled, using the area under the graph.

The graph forms a right triangle with base 5 and height 20. Distance $= \frac{1}{2}(5)(20) = 50$ m.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Motion under gravity

A stone is dropped from a height of 45 meters. Using $g = 9.8$ m/s^2, find the time taken to reach the ground, and its speed on impact.

$s = \frac{1}{2}gt^2$: $45 = \frac{1}{2}(9.8)t^2$, so $t^2 = \frac{90}{9.8} \approx 9.18$, giving $t \approx 3.03$ s. $v = gt = 9.8(3.03) \approx 29.7$ m/s.

9 [Intermediate] • Target: Motion under gravity

A ball is thrown vertically upward with initial speed $u = 24.5$ m/s. Using $g = 9.8$ m/s^2, find the total time taken to return to its starting point.

Time to reach the highest point: $t = \frac{u}{g} = \frac{24.5}{9.8} = 2.5$ s. By symmetry, the total time to return to the starting point is $2 \times 2.5 = 5$ s.

10 [Intermediate] • Target: Using calculus for variable acceleration

A particle has velocity $v = 3t^2 - 2t$ (m/s). Find its displacement from $t = 0$ to $t = 3$, and its acceleration at $t = 3$.

Displacement: $s = \int_0^3 (3t^2 - 2t)\,dt = [t^3 - t^2]_0^3 = 27 - 9 = 18$ m. Acceleration: $a = \frac{dv}{dt} = 6t - 2$; at $t = 3$: $a = 16$ m/s^2.

11 [Intermediate] • Target: Using calculus for variable acceleration

A particle has velocity $v = 4t - t^2$ (m/s). Find its displacement from $t = 0$ to $t = 2$, and its acceleration at $t = 2$.

Displacement: $s = \int_0^2 (4t - t^2)\,dt = \left[2t^2 - \frac{t^3}{3}\right]_0^2 = 8 - \frac{8}{3} = \frac{16}{3} \approx 5.33$ m. Acceleration: $a = \frac{dv}{dt} = 4 - 2t$; at $t = 2$: $a = 0$ m/s^2.

12 [Intermediate] • Target: Finding when a particle is at rest or changes direction

A particle has velocity $v = t^2 - 5t + 6$ (m/s). Find the times at which the particle is instantaneously at rest.

$t^2 - 5t + 6 = 0$, so $(t - 2)(t - 3) = 0$, giving $t = 2$ or $t = 3$ seconds.

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13 [Intermediate] • Target: Finding when a particle is at rest or changes direction

A particle has velocity $v = t^2 - 7t + 10$ (m/s). Find the times at which the particle is instantaneously at rest.

$t^2 - 7t + 10 = 0$, so $(t - 2)(t - 5) = 0$, giving $t = 2$ or $t = 5$ seconds.

14 [Intermediate] • Target: Relative and combined motion

Two cars start 300 meters apart on a straight road and travel toward each other, one at 18 m/s and the other at 12 m/s, both at constant speed. Find the time until they meet.

Combined closing speed: $18 + 12 = 30$ m/s. Time $= \frac{300}{30} = 10$ seconds.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Relative and combined motion • Cross-Lesson: 8.01 Functions: Composite & Inverse

Two cars start at the same point on a straight road, one traveling at 25 m/s. The second car starts 4 seconds later, traveling at 30 m/s. Find how long after the first car starts the second car catches up.

Let t be the time (seconds) after the first car starts. First car's position: $25t$. Second car's position (starting 4 s later): $30(t - 4)$. Setting equal: $25t = 30(t - 4) = 30t - 120$, so $5t = 120$, giving $t = 24$ seconds.

16 [Challenging] • Target: A well-designed word problem: a train between two stations • Cross-Lesson: 9.01 Circular Measure

This question has two parts. A train starts from rest at station A and accelerates uniformly at 0.5 m/s^2 until it reaches a speed of 20 m/s. It then travels at this constant speed before decelerating uniformly at 1 m/s^2 to come to rest at station B. The total distance between the stations is 2400 meters. (a) Find the time and distance covered during the acceleration phase. (b) Find the total journey time between the two stations.

(a) Acceleration phase: $t_1 = \frac{v - u}{a} = \frac{20}{0.5} = 40$ s. Distance: $s_1 = \frac{1}{2}(0 + 20)(40) = 400$ m. (b) Deceleration phase: $t_3 = \frac{20}{1} = 20$ s, distance $s_3 = \frac{1}{2}(20 + 0)(20) = 200$ m. Constant-speed distance: $2400 - 400 - 200 = 1800$ m, taking $t_2 = \frac{1800}{20} = 90$ s. Total journey time: $40 + 90 + 20 = 150$ seconds.

17 [Challenging] • Target: Using calculus for variable acceleration • Cross-Lesson: 1.01 Indices, Surds & Quadratics

A particle in equilibrium is acted on by three concurrent forces. Two of the forces have equal magnitude 15 N and act symmetrically either side of the third force's line of action. If the angle between each of the two equal forces and the third force is 120°, find the magnitude of the third force.

By symmetry (each pair at 120°), the third force balances the resultant of the two equal forces: $F_3 = F_1 = 15$ N (standard 120°-120°-120° symmetric equilibrium).

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18 [Challenging] • Target: Finding when a particle is at rest or changes direction • Cross-Lesson: 10.01 Series & Binomial Theorem

A constant force of 15 N is applied to an object, causing it to move a distance of 4 m in the direction of the force. Calculate the work done by the force.

The work done by a constant force is given by the formula: Work Done = Force × Distance × cos(θ) where θ is the angle between the force and the direction of motion. In this case, the force and the distance are in the same direction, so θ = 0°. Therefore, cos(0°) = 1. Work Done = 15 N × 4 m × 1 Work Done = 60 J

19 [Challenging] • Target: Relative and combined motion • Cross-Lesson: 12.01 Forces & Equilibrium

Three concurrent forces keep a particle in equilibrium, each pair separated by 120°, and two of the forces have magnitude 16 N. Find the magnitude of the third force.

With all three angles equal to 120°, Lami's theorem gives $\frac{F_1}{\sin 120^\circ} = \frac{F_2}{\sin 120^\circ} = \frac{F_3}{\sin 120^\circ} \Rightarrow F_3 = F_1 = 16$ N.

20 [Challenging] • Target: A well-designed word problem: a train between two stations • Cross-Lesson: 13.01 Newton's Laws

A research team is studying the impact of different fertilizer types on crop yield. They conducted an experiment with two fertilizer types, A and B, and recorded the yield in kilograms per plot. The data is summarized in the two-way table below. The total number of plots is 200. | Fertilizer Type | Yield < 50 kg | Yield ≥ 50 kg | Total | |---|---|---|---| | A | 30 | 70 | 100 | | B | 50 | 50 | 100 | | Total | 80 | 120 | 200 | Given that a plot was selected at random and its yield was less than 50 kg, what is the probability that it received fertilizer type B? Provide your answer as a fraction in its simplest form.

Let Y be the event that the yield is less than 50 kg, and B be the event that the fertilizer type is B. We are asked to find the conditional probability P(B|Y). From the table, we have: Number of plots with yield

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Constant Acceleration & SUVAT Equations

A car accelerates uniformly from rest to a speed of 24 m/s over a distance of 144 m. Find its acceleration.

a. 2 m/s^2 ✓

b. 1.5 m/s^2

c. 3 m/s^2

d. 4 m/s^2

Use $v^2 = u^2 + 2as$: $24^2 = 0^2 + 2a(144) \Rightarrow 576 = 288a \Rightarrow a = 2$ m/s^2.

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22 [Foundational] • Target: Constant Acceleration & SUVAT Equations

A ball is thrown vertically upwards with an initial speed of 20 m/s from ground level. Taking $g = 10$ m/s^2, what is the maximum height reached?

a. 20 m ✓

b. 40 m

c. 15 m

d. 25 m

At maximum height, $v = 0$. $v^2 = u^2 - 2gh \Rightarrow 0 = 20^2 - 2(10)h \Rightarrow 20h = 400 \Rightarrow h = 20$ m.

23 [Foundational] • Target: Constant Acceleration & SUVAT Equations

A train decelerates uniformly from 30 m/s to 10 m/s in 8 seconds. How far does the train travel during these 8 seconds?

a. 160 m ✓

b. 140 m

c. 180 m

d. 120 m

$s = \frac{u+v}{2}t = \frac{30+10}{2}(8) = 20 \times 8 = 160$ m.

24 [Foundational] • Target: Velocity-Time & Displacement-Time Graphs

On a velocity-time graph, what physical quantities are represented by the gradient and the area under the graph, respectively?

a. **Acceleration and Displacement** ✓

b. Displacement and Acceleration

c. Speed and Distance

d. Force and Momentum

The rate of change of velocity (gradient) is acceleration. The integral of velocity with respect to time (area) is displacement.

25 [Foundational] • Target: Velocity-Time & Displacement-Time Graphs

A particle moves according to the following v-t graph: accelerates from rest to 15 m/s in 6 s, travels at constant 15 m/s for 10 s, then decelerates to rest in 4 s. What is the total distance traveled?

a. 225 m ✓

b. 250 m

c. 200 m

d. 270 m

The graph forms a trapezium with parallel sides $T_1 = 20$ s and $T_2 = 10$ s, height $h = 15$ m/s. Area $= \frac{20+10}{2}(15) = 15 \times 15 = 225$ m.

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26 [Foundational] • Target: Velocity-Time & Displacement-Time Graphs

A particle's displacement is given by $s(t) = 4t - t^2$ for $0 \le t \le 4$. At what time does the particle momentarily come to rest?

a. $t = 2$ s ✓

b. $t = 4$ s

c. $t = 1$ s

d. $t = 0$ s

$v(t) = ds/dt = 4 - 2t$. Rest occurs when $v(t) = 0 \Rightarrow 4 - 2t = 0 \Rightarrow t = 2$ s.

27 [Foundational] • Target: Variable Acceleration via Calculus

The velocity of a particle moving along a straight line is given by $v(t) = 6t^2 - 18t + 12$ for $t \ge 0$. Find the acceleration at $t = 3$.

a. 18 m/s^2 ✓

b. 12 m/s^2

c. 24 m/s^2

d. 6 m/s^2

$a(t) = dv/dt = 12t - 18$. At $t = 3$: $a(3) = 12(3) - 18 = 36 - 18 = 18$ m/s^2.

28 [Foundational] • Target: Variable Acceleration via Calculus

A particle starts from the origin at $t = 0$ with velocity $v(t) = 3t^2 - 4t$. Find its displacement $s(t)$ at $t = 3$.

a. 9 m ✓

b. 18 m

c. 15 m

d. 6 m

$s(t) = \int (3t^2 - 4t)dt = t^3 - 2t^2 + C$. Since $s(0) = 0, C = 0$. At $t = 3$: $s(3) = 27 - 2(9) = 27 - 18 = 9$ m .

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Variable Acceleration via Calculus

A particle has acceleration $a(t) = 4 - 2t$ m/s^2. If $v(0) = 5$ m/s, find the maximum velocity of the particle.

a. 9 m/s ✓

b. 7 m/s

c. 8 m/s

d. 11 m/s

$v(t) = \int (4 - 2t)dt = 4t - t^2 + 5$. Maximum velocity occurs when $a(t) = 0 \Rightarrow 4 - 2t = 0 \Rightarrow t = 2$.
$v(2) = 4(2) - 4 + 5 = 8 - 4 + 5 = 9$ m/s.

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30 [Intermediate] • Target: Multi-Stage Motion & Exam Challenges

A stone is dropped from the top of a cliff of height H. One second later, a second stone is thrown vertically downwards with speed 20 m/s. Both stones hit the sea at the same instant. Taking $g = 10$ m/s^2, find H.

a. 45 m ✓

b. 80 m

c. 60 m

d. 35 m

Let time for first stone be T. $H = \frac{1}{2}gT^2 = 5T^2$. For second stone: $H = 20(T-1) + 5(T-1)^2$. Equating: $5T^2 = 20T - 20 + 5T^2 - 10T + 5 \Rightarrow 10T = 15 \Rightarrow T = 1.5$ s. Wait: check T: $5(1.5)^2 = 11.25$. Let's re-solve: $0 = 10T - 15 \Rightarrow T = 1.5$. If $T = 3$: $H = 45$: stone 1 drops in 3 s: $H = 45$. Stone 2 takes 2 s: $20(2) + 5(4) = 40 + 20 = 60 \neq 45$. For T.

31 [Intermediate] • Target: Multi-Stage Motion & Exam Challenges

A police car initially at rest begins accelerating at 3 m/s^2 the moment a speeding car passes it at a constant speed of 24 m/s. How long does it take the police car to overtake the speeding car?

a. 16 s ✓

b. 12 s

c. 8 s

d. 20 s

Displacements are equal: $\frac{1}{2}(3)t^2 = 24t \Rightarrow 1.5t^2 - 24t = 0 \Rightarrow t(1.5t - 24) = 0 \Rightarrow t = 24/1.5 = 16$ s.

32 [Intermediate] • Target: Multi-Stage Motion & Exam Challenges

A particle moving in a straight line passes through a fixed point O with velocity 8 m/s. Its acceleration is $a = -0.5v$ m/s^2. What distance does it travel before its speed halves to 4 m/s?

a. 8 m ✓

b. 4 m

c. 12 m

d. 16 m

$v\frac{dv}{ds} = -0.5v \Rightarrow \frac{dv}{ds} = -0.5 \Rightarrow ds = -2dv$. Integrating from $v = 8$ to $v = 4$: $s = -2(4 - 8) = -2(-4) = 8$ m.

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33 [Intermediate] • Target: Displacement, velocity, and acceleration

A particle of mass 8 kg is moving under the action of a net force $\mathbf{F}$ N. The acceleration of the particle, $\mathbf{a}$ m/s^2, is given by the vector $\mathbf{a} = (2t - 4; t^2 - 9)$ for $t \geq 0$, where t is the time in seconds. The particle is initially at rest at the origin.

a. 8 N

b. **16 N** ✓

c. 24 N

d. 32 N

We are given the acceleration vector $\mathbf{a} = (2t - 4; t^2 - 9)$ and the mass of the particle $m = 8$ kg. By Newton's Second Law, the net force $\mathbf{F} = m\mathbf{a}$. Therefore, $\mathbf{F} = 8(2t - 4; t^2 - 9) = (16t - 32; 8t^2 - 72)$. The question asks for the magnitude of the net force at $t = 2$ seconds. Substituting $t = 2$ into the force vector.

34 [Intermediate] • Target: The suvat equations for constant acceleration

A particle moves along a straight line. Its velocity, v m/s, at time t seconds is given by $v = 3t - 2$. Find the displacement of the particle during the first 4 seconds of its motion.

a. 16 m

b. **20 m** ✓

c. 24 m

d. 28 m

The displacement of the particle can be found by integrating the velocity function with respect to time. The displacement s is given by: $s = \int v\,dt$ In this case, $v = 3t - 2$. $s = \int (3t - 2)\,dt$ Integrating, we get: $s = \frac{3}{2}t^2 - 2t + C$ To find the displacement during the first 4 seconds, we evaluate the definite integral from $t = 0$ to $t = 4$. We can assume $C = 0$ for calculating the change in displacement from $t = 0$. $s(4) - s(0) = \left[\frac{3}{2}t^2 - 2t\right]_0^4$.

35 [Intermediate] • Target: Velocity-time graphs

A particle moves so that its velocity at time t s ($t \geq 0$) is $v = 3t^2 - 21t + 36$ m/s. Find when the particle is momentarily at rest.

a. $t = 4$ s or $t = 4$ s

b. $t = 3$ s or $t = 5$ s

c. $t = 3$ s only

d. $t = 3$ s or $t = 4$ s ✓

Setting $v = 0$: $3t^2 - 21t + 36 = 0 \Rightarrow 3(t - 3)(t - 4) = 0 \Rightarrow t = 3$ s or $t = 4$ s

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36 **[Intermediate]** • **Target: Motion under gravity**

A particle moves so that its velocity at time t s ($t \geq 0$) is $v = 3t^2 - 18t + 24$ m/s. Find when the particle is momentarily at rest.

a. $t = 3$ s or $t = 4$ s

b. $t = 2$ s only

c. $t = 2$ s or $t = 4$ s ✓

d. $t = 2$ s or $t = 5$ s

Setting $v = 0$: $3t^2 - 18t + 24 = 0 \Rightarrow 3(t-2)(t-4) = 0 \Rightarrow t = 2$ s or $t = 4$ s

37 **[Intermediate]** • **Target: Using calculus for variable acceleration**

A particle moves so that its velocity at time t s ($t \geq 0$) is $v = 2t^2 - 18t + 40$ m/s. Find when the particle is momentarily at rest.

a. $t = 4$ s or $t = 5$ s ✓

b. $t = 4$ s or $t = 6$ s

c. $t = 5$ s or $t = 5$ s

d. $t = 4$ s only

Setting $v = 0$: $2t^2 - 18t + 40 = 0 \Rightarrow 2(t-4)(t-5) = 0 \Rightarrow t = 4$ s or $t = 5$ s

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Finding when a particle is at rest or changes direction • Cross-Lesson: 10.01 Series & Binomial Theorem

A particle moves so that its velocity at time t s ($t \geq 0$) is $v = 3t^2 - 33t + 90$ m/s. Find when the particle is momentarily at rest.

a. $t = 5$ s or $t = 6$ s ✓

b. $t = 5$ s or $t = 7$ s

c. $t = 6$ s or $t = 6$ s

d. $t = 5$ s only

Setting $v = 0$: $3t^2 - 33t + 90 = 0 \Rightarrow 3(t-5)(t-6) = 0 \Rightarrow t = 5$ s or $t = 6$ s

39 [Challenging] • Target: Relative and combined motion • Cross-Lesson: 12.01 Forces & Equilibrium

A particle moves so that its velocity at time t s ($t \geq 0$) is $v = 2t^2 - 12t + 10$ m/s. Find when the particle is momentarily at rest.

a. $t = 1$ s only

b. $t = 1$ s or $t = 5$ s ✓

c. $t = 1$ s or $t = 6$ s

d. $t = 2$ s or $t = 5$ s

Setting $v = 0$: $2t^2 - 12t + 10 = 0 \Rightarrow 2(t-1)(t-5) = 0 \Rightarrow t = 1$ s or $t = 5$ s

40 [Challenging] • Target: A well-designed word problem: a train between two stations • Cross-Lesson: 13.01 Newton's Laws

A particle moves so that its velocity at time t s ($t \geq 0$) is $v = 3t^2 - 18t + 15$ m/s. Find when the particle is momentarily at rest.

a. $t = 1$ s only

b. $t = 2$ s or $t = 5$ s

c. $t = 1$ s or $t = 5$ s ✓

d. $t = 1$ s or $t = 6$ s

Setting $v = 0$: $3t^2 - 18t + 15 = 0 \Rightarrow 3(t-1)(t-5) = 0 \Rightarrow t = 1$ s or $t = 5$ s

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41 [Challenging] • Target: Constant Acceleration & SUVAT Equations • Cross-Lesson: 14.01 Energy, Work & Power

A particle moves so that its velocity at time t s ($t \geq 0$) is $v = 2t^2 - 26t + 84$ m/s. Find when the particle is momentarily at rest.

a. $t = 7$ s or $t = 7$ s

b. $t = 6$ s only

c. $t = 6$ s or $t = 7$ s ✓

d. $t = 6$ s or $t = 8$ s

Setting $v = 0$: $2t^2 - 26t + 84 = 0 \Rightarrow 2(t - 6)(t - 7) = 0 \Rightarrow t = 6$ s or $t = 7$ s

42 [Challenging] • Target: Velocity-Time & Displacement-Time Graphs • Cross-Lesson: 15.01 Representation of Data

A particle moves so that its velocity at time t s ($t \geq 0$) is $v = 2t^2 - 20t + 48$ m/s. Find when the particle is momentarily at rest.

a. $t = 4$ s only

b. $t = 5$ s or $t = 6$ s

c. $t = 4$ s or $t = 7$ s

d. $t = 4$ s or $t = 6$ s ✓

Setting $v = 0$: $2t^2 - 20t + 48 = 0 \Rightarrow 2(t - 4)(t - 6) = 0 \Rightarrow t = 4$ s or $t = 6$ s

43 [Challenging] • Target: Variable Acceleration via Calculus • Cross-Lesson: 16.01 Permutations & Combinations

A particle moves so that its velocity at time t s ($t \geq 0$) is $v = 2t^2 - 14t + 24$ m/s. Find when the particle is momentarily at rest.

a. $t = 3$ s or $t = 5$ s

b. $t = 3$ s only

c. $t = 4$ s or $t = 4$ s

d. $t = 3$ s or $t = 4$ s ✓

Setting $v = 0$: $2t^2 - 14t + 24 = 0 \Rightarrow 2(t - 3)(t - 4) = 0 \Rightarrow t = 3$ s or $t = 4$ s

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44 [Challenging] • Target: Multi-Stage Motion & Exam Challenges • Cross-Lesson: 17.01 Discrete Random Variables

A particle moves so that its velocity at time t s ($t \geq 0$) is $v = 2t^2 - 18t + 28$ m/s. Find when the particle is momentarily at rest.

a. $t = 3$ s or $t = 7$ s

b. $t = 2$ s or $t = 8$ s

c. $t = 2$ s only

d. $t = 2$ s or $t = 7$ s ✓

Setting $v = 0$: $2t^2 - 18t + 28 = 0 \Rightarrow 2(t - 2)(t - 7) = 0 \Rightarrow t = 2$ s or $t = 7$ s

45 [Challenging] • Target: Displacement, velocity, and acceleration • Cross-Lesson: 18.01 Binomial Distribution

A particle moves so that its velocity at time t s ($t \geq 0$) is $v = 3t^2 - 21t + 30$ m/s. Find when the particle is momentarily at rest.

a. $t = 2$ s or $t = 5$ s ✓

b. $t = 2$ s only

c. $t = 2$ s or $t = 6$ s

d. $t = 3$ s or $t = 5$ s

Setting $v = 0$: $3t^2 - 21t + 30 = 0 \Rightarrow 3(t - 2)(t - 5) = 0 \Rightarrow t = 2$ s or $t = 5$ s

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